

THEORY AND METHODS

CALCULATING CENTRAL AND NON-CENTRAL
F PROBABILITIES

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Summary: A simple and efficient method of calculating F probabilities is presented.

1. INTRODUCTION

It is generally thought that the calculation of F probabilities is difficult. This apparent difficulty is illustrated in a recent description of a solution to the problem, published by Singh and Relyea (1992). Almost all the algorithms published to date distinguish a number of special cases and invoke complicated formulas for each case. This compartmentalization results in a degree of recalculation and such algorithms also often require the use of an existing routine for a subsidiary part of the calculation. For examples exhibiting these complication, see Majumder and Bhattacharjee (1973), Klotz and Meyer (1985), Narula and Weistroffer (1986), Guirguis (1990) and Posten (1993). For a completely different (but not easier) approach to that used in these references and this note, see Rust and Voit (1990). The purpose of this note is to display a simple self-contained method for the calculation of central F probabilities which uses recurrence relations and which greatly simplifies the calculation of non-central F probabilities.

2. THE METHOD

A general formula for the calculation of F probabilities is

$$(1) \quad P[F(m, n, \lambda) \geq f] = \sum_{r=0}^{\infty} \frac{e^{-\frac{1}{2}\lambda} (\frac{1}{2}\lambda)^r}{r!} I_x(a, b+r)$$

in which $a = n/2$, $b = m/2$ and $x = n/(n+mf)$, while $I_x(a, b)$ is the incomplete beta function defined in 26.5.1 of Abramowitz and Stegun (1975). The degrees of freedom, m and n , are (for simplicity) required to be integers in this note; λ is the non-centrality parameter of the distribution.

The formula (1) simplifies to $I_x(a, b)$ when $\lambda = 0$. A general routine for calculating F probabilities based on (1) should therefore start by calculating this quantity. For non-zero λ , having calculated $I_x(a, b)$, a recurrence relation (easily derived from 26.5.2 and 26.5.16 in Abramowitz and Stegun) can be used to calculate subsequent terms in the expansion (1), Lenth (1987). The relation is

$$(2) \quad I_x(a, b+1) = I_x(a, b) + \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b+1)} x^a(1-x)^b.$$

Writing $T_x(a, b)$ for the second term in (2), note that

$$(3) \quad T_x(a, b+1) = T_x(a, b) \frac{(a+b)}{(b+1)}(1-x)$$

and hence that if the calculation of $I_x(a, b)$ was arranged to be by way of (2) and (3) then (by continuing to use 3) it will be a simple extension to continue calculating (1) for non-zero λ . Thus, we must begin by describing the calculation of $I_x(a, b)$.

The need to recognize special cases and deal with them separately cannot be excluded in the calculation of $I_x(a, b)$, but in the method described here compartmentalization is restricted almost entirely to the initializations. There are three special cases.

Case I: m even

$$(4) \quad I_x(a, 1) = x^a$$

$$(5) \quad A = a, \quad B = 1 \quad \text{and} \quad T_x(a, 1) = ax^a(1-x)$$

This initialization is followed by repeated use of (2) and (3) ($b-B$ times) to yield $I_x(a, b)$. Note that one always ends an iteration with the value of $T_x(a, b)$ for the next iteration available. Furthermore, if both m and n are even and n is less than m then there is advantage in this case of calculating $I_x(a, b)$ according to the method described for Case II below.

Case II: m odd, n even

Since $I_x(a, b) = 1 - I_{1-x}(b, a)$, in which $a = n/2$ is integer, it follows that (2) may also be applied in this case. By contrast, the direct calculation of $I_x(1, b)$, starting with $I_x(1, b)$ and iterating with 25.5.16 in Abramowitz and Stegun, is to be avoided here because the subtraction in 25.5.16 can lead to cancellation (cf. Randall and Rayner, 1990) and hence a negative value for $I_x(a, b)$. A second advantage of the approach in this case is that the calculation is identical to that described for Case I, except that the roles of a and b , and of x and $1-x$, are reversed. After the initializations

$$(6) \quad I_{1-x}(b, 1) = (1-x)^b$$

$$(7) \quad A = 1, \quad B = b \quad \text{and} \quad T_{1-x}(b, 1) = b(1-x)^b x$$

$a-A$ iterations follow (increasing A to a) to yield $I_{1-x}(b, a)$ and $T_{1-x}(B, A)$. The latter is converted to $T_x(A, B)$ by multiplying with A/B .

Case III: m and n both odd

This case is initialized with

$$(8) \quad I_{1-x}(B, A) = \frac{2}{\pi} \arctan \sqrt{\frac{1-x}{x}}$$

$$(9) \quad A = \frac{1}{2}, \quad B = \frac{1}{2} \quad \text{and} \quad T_{1-x}(B, A) = \frac{2}{\pi} \sqrt{(1-x)x}$$

This is followed by the iteration described in Case II, which passes through $I_{1-x}(B, a)$ and $T_{1-x}(B, a)$ to $I_x(a, B)$ and $T_x(a, B)$. The iteration described in Case I then yields $I_x(a, b)$ and $T_x(a, b)$. Thus, apart from the initialization, the present case simply utilizes existing programming. Compare this to 26.6.8 in Abramowitz and Stegun, a popular method used in

conjunction with 26.6.4 and 26.6.5 for the calculation of central F probabilities.

The real advantage of the method of calculation of $I_x(a, b)$ described above is now clear. It yields $T_x(a, b)$ as a by-product and hence calculation of (1) may proceed employing (2) iteratively.

In the evaluation of (1), note that if λ is very large, special precautions must be used to prevent the calculation failing because of underflow in the evaluation of $e^{-\lambda}$ (Lee, 1988a, 1988b). Such underflow may explain why the VAX SAS version 6.07 function for the evaluation of non-central F probabilities (running under version 5.5-1 of VMS on a VAX 6000/410) fails when used to calculate

$$(10) \quad P = \text{PROBF}(x, \text{ndf}, \text{ddf}, \text{nc})$$

when $x = 10 \times (1-0.01)/0.01$, $\text{ndf} = 1$, $\text{ddf} = 12$ and $\text{nc} = 12 \times 193$, as required in the case of the entry in the bottom left-hand corner of table 1 in Chow and Shao (1990). The function does not fail for the same value of x but lower values of nc (the non-centrality parameter), for example in the case of the entry in Chow and Shao's table for which $\text{nc} = 2 \times 132$.

Failure of the calculation can be avoided by factoring $e^{-\lambda}$ into a series of factors equal to the current partial sum in the summation of (1) omitting $e^{-\lambda}$ (so that the r -th factor transforms the current partial sum to unity), until the remaining factor is well below the threshold at which underflow occurs. At this point the remaining factor may be introduced into the calculation which is then completed in accordance with the literal interpretation of (1).

The algorithm described above has been programmed (in Pascal) and a listing is available from the author. It has been tested against the extensive tables given in Graybill (1976), and against SAS's PROBF function (except in cases of the failure of the latter). The principles in the algorithm may also be applied in the calculation of upper-tail non-central t probabilities, using equation (3) in Hawkins (1975) with parallel iterations on the odd-indexed and even-indexed terms of the series expansion.

3. CONCLUSION

It is shown that if non-central F (and t) probabilities are calculated with the aid of an incomplete beta function, having this function output $T_x(a, b)$ in addition to $I_x(a, b)$ (where the former is a by-product of the calculation of the latter) avoids the need to use subsidiary routines in the calculation.

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